

Normalization

Unnormalized Form

Stud_ID	Stud_Name	Stud_Email	Book_ID	Book_Title	Author_Name	Category_Name	Status_Name	Borrow_Date	Return_Date
1	Ahmed Hassan	ahmed@gmail.com	1	1984	George Orwell	Fiction	Available	2025-01-10	2025-01-20
1	Ahmed Hassan	ahmed@gmail.com	2	Animal Farm	George Orwell	Fiction	Available	2025-02-01	2025-02-10
2	Sara Ali	sara@gmail.com	2	Animal Farm	George Orwell	Fiction	Borrowed	2025-01-12	2025-01-22
3	Emad Essam	emad@gmail.com	4	Great Expectations	Jane Eyre	Charlotte Brontë	Available	2025-01-22	2025-01-30
4	Mohamed Adel	mohamed@gmail.com	10	Clean Code	Robert Martin	Programming	Available	2025-03-01	2025-03-10
4	Mohamed Adel	mohamed@gmail.com	11	Clean Architecture	Robert Martin	Programming	Available	2025-03-15	2025-03-25
5	Nour Ahmed	nour@gmail.com	12	Database Systems	Elmasri	Education	Borrowed	2025-03-05	2025-03-20
5	Nour Ahmed	nour@gmail.com	10	Clean Code	Robert Martin	Programming	Borrowed	2025-03-18	2025-03-30

Step 1: First Normal Form (1NF)

- 1- Each column must contain **atomic** (single, indivisible) values.
- 2- There should be **no repeating groups** of columns.

Students Table

Stud_ID	Stud_Name	Stud_Email
1	Ahmed Hassan	ahmed@gmail.com
2	Sara Ali	sara@gmail.com
3	Emad Essam	emad@gmail.com
4	Mohamed Adel	mohamed@gmail.com
5	Nour Ahmed	nour@gmail.com

Books Table

<u>Book_ID</u>	Book_Title	Author_Name	Category_Name
1	1984	George Orwell	Fiction
2	Animal Farm	George Orwell	Fiction
4	Great Expectations	Jane Eyre	Charlotte Brontë
10	Clean Code	Robert Martin	Programming
11	Clean Architecture	Robert Martin	Programming
12	Database Systems	Elmasri	Education

Borrow Table

<u>Stud_ID</u>	<u>Book_ID</u>	Status_Name	Borrow_Date	Return_Date
1	1	Available	2025-01-10	2025-01-20
1	2	Available	2025-02-01	2025-02-10
2	2	Borrowed	2025-01-12	2025-01-22
3	4	Available	2025-01-22	2025-01-30
4	10	Available	2025-03-01	2025-03-10
4	11	Available	2025-03-15	2025-03-25
5	12	Borrowed	2025-03-05	2025-03-20
5	10	Borrowed	2025-03-18	2025-03-30

Problems:

- Data redundancy still exists.
 - Non-key attributes depend on other non-key attributes.
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Step 2: Second Normal Form (2NF)

- 1- The table must already be in **1NF**.
- 2- There must be **no partial dependency** of any non-prime attribute on any candidate key.

STUDENT

<u>Stud_ID</u>	Stud_Name	Stud_Email
1	Ahmed Hassan	ahmed@gmail.com
2	Sara Ali	sara@gmail.com
3	Emad Essam	emad@gmail.com
4	Mohamed Adel	mohamed@gmail.com
5	Nour Ahmed	nour@gmail.com

BOOK

<u>Book_ID</u>	Book_Title	Author_Name	Category_Name
1	1984	George Orwell	Fiction
2	Animal Farm	George Orwell	Fiction
4	Great Expectations	Jane Eyre	Charlotte Brontë
10	Clean Code	Robert Martin	Programming
11	Clean Architecture	Robert Martin	Programming
12	Database Systems	Elmasri	Education

BORROW

<u>Stud_ID</u>	<u>Book_ID</u>	Status_Name	Borrow_Date	Return_Date
1	1	Available	2025-01-10	2025-01-20
1	2	Available	2025-02-01	2025-02-10
2	2	Borrowed	2025-01-12	2025-01-22
3	4	Available	2025-01-22	2025-01-30
4	10	Available	2025-03-01	2025-03-10
4	11	Available	2025-03-15	2025-03-25
5	12	Borrowed	2025-03-05	2025-03-20
5	10	Borrowed	2025-03-18	2025-03-30

Problem:

- Transitive dependency exists in BOOK table.

Step 3: Third Normal Form (3NF)

- 1- The table must already be in **2NF**.
- 2- There must be **no transitive dependency** of any non-prime attribute on the candidate key.

STUDENT

<u>Stud_ID</u>	Stud_Name	Stud_Email
1	Ahmed Hassan	ahmed@gmail.com
2	Sara Ali	sara@gmail.com
3	Emad Essam	emad@gmail.com
4	Mohamed Adel	mohamed@gmail.com
5	Nour Ahmed	nour@gmail.com

AUTHOR

<u>Author_ID</u>	Author_Name
1	George Orwell
2	Charlotte Brontë
3	Robert Martin
4	Elmasri

CATEGORY

<u>Category_ID</u>	Category_Name
1	Fiction
2	Programming
3	Education

BOOK

<u>Book_ID</u>	Book_Title	Author_ID	Category_ID
1	1984	1	1
2	Animal Farm	1	1
4	Great Expectations	2	1
10	Clean Code	3	2
11	Clean Architecture	3	2
12	Database Systems	4	3

STATUS

<u>Status_ID</u>	Status_Name
1	Available
2	Borrowed

BORROW

<u>Borrow_ID</u>	Stud_ID	Book_ID	Status_ID	Borrow_Date	Return_Date
1	1	1	1	2025-01-10	2025-01-20
2	1	2	1	2025-02-01	2025-02-10
3	2	2	2	2025-01-12	2025-01-22
4	3	4	1	2025-01-22	2025-01-30
5	4	10	1	2025-03-01	2025-03-10
6	4	11	1	2025-03-15	2025-03-25
7	5	12	2	2025-03-05	2025-03-20
8	5	10	2	2025-03-18	2025-03-30